

Bill of Materials

SMD PCB Circuits — Pick & Place Demonstration

All components available in Switzerland

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Circuit 1 — Flashing LED (NE555 Astable)

Description

A minimal SMD LED flasher based on the NE555 timer IC in astable mode. Flashes a single red LED at a fixed frequency set by resistors and capacitor. Power via USB Micro-B connector. Repository: github.com/ItsKarlito/flashing-led

Component List

- 1x NE555DR — Timer IC, SOIC-8
- 1x LED, Red, SMD 0805
- 1x USB Micro-B Connector, Female, SMD
- 1x Capacitor 1 uF, SMD Electrolytic
- 1x Resistor 100 kOhm, SMD 0805
- 1x Resistor 470 kOhm, SMD 0805
- 1x Resistor 270 kOhm, SMD 0805

Circuit 2 — LED Chaser / Knight Rider (NE555 + CD4017)

Description

A 10-LED running light chaser. The NE555 generates a clock in astable mode, driving the CD4017 decade counter. Each output lights one LED in sequence. Power supply: 9V DC. Reference: [PCBWay — NE555 LED Chaser Board](#)

Component List

- 1x NE555DR — Timer IC, SOIC-8
- 1x CD4017BM96 — Decade Counter IC, SOIC-16
- 10x LED, Red, SMD 0805
- 1x Resistor 10 kOhm, SMD 0805
- 10x Resistor 220 Ohm, SMD 0805
- 1x Capacitor 10 nF, SMD Ceramic 0805
- 1x Capacitor 10 uF, SMD Ceramic 0805

Combined Bill of Materials

NE555DR Timer IC	2	SOIC-8	mouser.ch	Search: NE555DR
CD4017BM96 Decade Counter	1	SOIC-16	mouser.ch	Search: CD4017BM96
VLMS1300-GS08 LED, Red	11	SMD 0805	mouser.ch	Search: VLMS1300
RND 1550805S8F2200T5E Resistor 220 Ohm	10	SMD 0805, 1%	distrelec.ch	Art. 30056759
RND 1550805S8F1002T5E Resistor 10 kOhm	1	SMD 0805, 1%	distrelec.ch	Art. 30056718
RC0805FR-07100KL Resistor 100 kOhm	1	SMD 0805, 1%	mouser.ch	Search: RC0805FR-07100KL
RC0805FR-07270KL Resistor 270 kOhm	1	SMD 0805, 1%	mouser.ch	Search: RC0805FR-07270KL
RC0805FR-07470KL Resistor 470 kOhm	1	SMD 0805, 1%	mouser.ch	Search: RC0805FR-07470KL
C0805C103K5RACTU Cap 10 nF	1	SMD 0805, X7R, 50V	distrelec.ch	Art. 16572413
EEE-FK1H010P Cap 1 uF	1	SMD Electrolytic	mouser.ch	Search: EEE-FK1H010P
CC0805KKX5R8BB106 Cap 10 uF	1	SMD 0805, 25V	distrelec.ch	Art. 30288132
47346-0001 (Molex) USB Micro-B	1	Female, SMD, 5-pin	mouser.ch	Search: 47346-0001
